TEXORA · LECTURE SHEET

Solvent Dyeing and Modern Techniques of Dyeing

From perchloroethylene baths to supercritical carbon dioxide — why dyeing without water was tried, why the first version faded, and which modern routes now carry the idea forward.

WP-4203 · SPECIAL WET PROCESSING · TOPIC 2 · SPECIALIZATION — WET PROCESSING

1. Why look beyond water at all?

Water is cheap, safe and an excellent dyeing medium — which is why it dominates. Its costs are hidden: heating large baths, drying wet fabric, hydrolysing reactive dyes, generating coloured effluent that must be treated, and swelling fibres in ways that complicate some processes. Every technology in this sheet is an attempt to keep water's advantages (dye transport, levelness, fibre access) while cutting one or more of those costs. They fall into three families: **replace water with another solvent** (solvent dyeing, supercritical CO_2), **use drastically less water** (foam dyeing — Topic 1; air-assisted and low-liquor systems), and **change how colour is delivered or how energy enters the bath** (digital inkjet, ultrasound, microwave, plasma assistance).

2. Solvent dyeing — principle

In **solvent dyeing** the dye bath is an organic solvent (or a solvent-rich mixture) instead of water. The historically important solvents were chlorinated hydrocarbons — above all **perchloroethylene** (tetrachloroethylene, "perc", also familiar as dry-cleaning solvent), with trichloroethylene and 1,1,1-trichloroethane also studied. The principle has four steps:

1. **Dissolution / dispersion:** dyes suited to the solvent — mainly disperse dyes for polyester and other synthetics, and selected solvent-soluble dyes — are dissolved or finely dispersed in the solvent. Fibres such as polyester swell in hot chlorinated solvent, opening the structure to dye at lower effective temperatures than aqueous dyeing.
2. **Exhaustion:** fabric circulates in the solvent bath in a sealed machine resembling a dry-cleaning machine enlarged for textiles. Dye partitions from solvent into fibre.
3. **Solvent removal:** solvent is drained and the fabric dried by hot air / vacuum in the closed machine; evaporated solvent is condensed.

4. **Recovery and reuse:** recovered solvent is distilled, purified and returned to the bath. The entire economics and environmental case stand or fall on recovery efficiency — solvent must be recycled at very high percentage, batch after batch.

Demonstrated advantages in its era: no aqueous effluent from the dye bath itself; no drying of water (solvent evaporates faster, with lower latent heat per kg than water); excellent results on polyester and on water-sensitive goods; solvent also scours as it dyes, combining preparation and colouration. Research systems also explored solvent-assisted dyeing of wool and blends, and solvent media for applying finishes — but cotton with its ionic reactive/vat chemistry, which *needs* water and alkali, never fitted the chlorinated-solvent route well. That fibre limitation mattered: the technology served synthetics best, just as aqueous polyester dyeing (high-temperature and carrier methods) was becoming good and cheap.

3. Why solvent dyeing declined

- **Capital and operating cost:** sealed, explosion- and emission-proof machines, condensers and distillation plant cost far more than aqueous machines; solvent purchase and make-up for inevitable losses added running cost.
- **Recovery losses:** every percent of solvent lost is both a cost and an emission. Real plants could not sustain the near-total recovery the model required across loading, unloading and maintenance.
- **Health and environment:** chlorinated solvents came under increasing regulatory pressure — workplace exposure limits, groundwater contamination risk, and classification concerns for perchloroethylene. Venting or spilling a chlorinated solvent is a far more serious event than spilling water.
- **Competition improved:** aqueous dyeing did not stand still — low-liquor jet machines, better disperse dyes, foam and other low-pickup routes captured much of the saving without the solvent burden.

The lesson carried into modern work: a waterless process only wins if its medium is **cheap, non-toxic, and recoverable in a simple closed loop**. That is exactly the design brief that supercritical CO_2 satisfies — the solvent changed, the idea survived.

4. Supercritical CO_2 dyeing

4.1 The medium

Carbon dioxide above its critical point — **critical temperature 31.1 °C, critical pressure 73.8 bar** — becomes a **supercritical fluid (scCO_2)**: density like a liquid (so it dissolves disperse dyes), diffusivity and low viscosity like a gas (so it penetrates fibre and yarn packages rapidly), and zero surface tension

(so it wets everything). Its solvent power is tunable with pressure: raise pressure, raise density, dissolve more dye.

CO_2 critical point: T_c = 31.1 °C | P_c = 73.8 bar | Typical dyeing: ~120 °C, ~250 bar (typical operating ranges)

4.2 The process loop

1. Liquid CO_2 from storage is compressed by a **pump** to dyeing pressure and heated above T_c.
2. It passes through a **dye vessel**, dissolving disperse dye (the same dye class used for aqueous polyester dyeing).
3. In the **dyeing autoclave**, fabric (typically polyester, as piece or beam) is held while dye-laden $scCO_2$ circulates; dye partitions into the fibre, levelled by the fluid's high diffusivity. No water, no dispersing agents beyond what the dye needs in CO_2, no reduction clearing afterwards.
4. On depressurisation through a **separator**, CO_2 drops its solubility, unused dye precipitates and is recovered, and clean CO_2 gas is condensed and **recycled** to storage — commercial designs recover the great majority of CO_2 each cycle.
5. The fabric comes out **dry**. There is no water to evaporate and no effluent stream from dyeing.

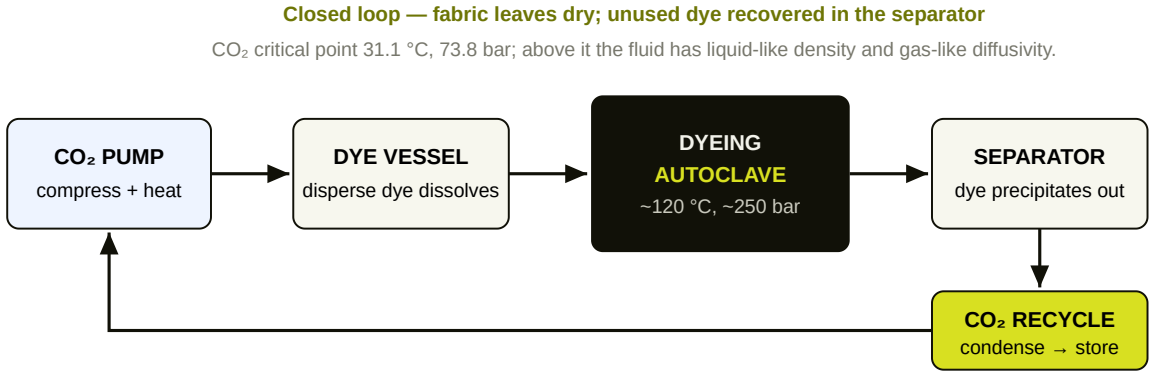


Figure 1 — Supercritical CO_2 dyeing loop. Pump → dye vessel → autoclave → separator → recycle; depressurisation both recovers the dye and regenerates clean CO_2.

Strengths: zero dyeing effluent; no drying energy for water; fast, level dyeing of polyester; dye and CO_2 both recycled; no after-clearing. **Limits:** high-pressure vessels are expensive and batch-sized; the proven fibre is polyester (and some synthetics/blends with special dyes) — hydrophilic fibres such as cotton lack dye systems that dissolve in CO_2, an active research area (functionalised dyes, co-solvents, reverse approaches); shade changes require careful cleaning to avoid cross-contamination. $scCO_2$ is therefore a commercial reality for polyester niches and a benchmark for what "waterless" must achieve, not yet a general replacement for aqueous dyeing.

5. Other modern techniques

5.1 Waterless / air dyeing concepts

"Air dyeing" in its commercial form is not dyeing in air as a solvent; it is a family of processes that transfer dye with the help of air flow and very small amounts of water — for example injecting dye solution into a high-velocity air stream that carries it through the fabric (the principle behind air-flow dyeing machines, which also cut liquor ratio in conventional exhaust dyeing to about 1:3–1:5). Marketing claims of ~95% water saving refer to specific comparisons; the engineering content is: transport by air instead of liquor, minimal bath, rapid exchange. Evaluate any "waterless" claim by asking what carries the dye, what fixes it, and what is washed off afterwards.

5.2 Digital inkjet printing as coloration

Inkjet is printing, but at production scale it is also a *coloration route* that competes with dyeing for patterned and short-run work: drop-on-demand heads jet dye-based inks (reactive for cotton, acid for silk/nylon, disperse for polyester — direct or transfer, pigment for universal use) only where the design needs colour. Water and chemical use per metre are a fraction of screen printing (no screens, no paste wash-off), changeover is a file, not a wash-down. Costs are ink price and pre-/post-treatment (fabric must still be prepared, steamed/washed for dye inks). See also Topic 3 on placement printing.

5.3 Ultrasound-assisted dyeing

Ultrasonic cavitation — collapsing micro-bubbles in the bath — creates micro-jets that de-agglomerate dye, degas fibre capillaries and accelerate diffusion to and into the fibre. Reported benefits: faster exhaustion, lower temperature or shorter time, improved levelness on some systems. Scale-up (uniform acoustic fields in large machines) and equipment cost have kept it at pilot/niche scale.

5.4 Microwave-assisted dyeing

Microwaves heat the water *inside* the bath and fibre volumetrically and rapidly, rather than heating vessel walls first. Laboratory and small-scale work shows faster heating, shorter dyeing cycles and good fixation for reactive and other systems. Uniformity at bulk scale (hot spots, penetration depth) is the engineering constraint.

5.5 Plasma-assisted dyeing

Atmospheric or low-pressure plasma (Topic 3) activates fibre surfaces — raising surface energy, etching nano-roughness, grafting polar groups — so subsequent aqueous dyeing wets faster, exhausts better, or works on difficult fibres (wool, polypropylene research). Plasma here is a *pre-treatment assistant* to dyeing, not the dyeing medium itself.

5.6 Foam dyeing (cross-reference)

Foam dyeing, covered fully in Topic 1, belongs in every modern-techniques comparison: it needs no exotic medium, retrofits near conventional lines, and cuts water and drying energy by applying dye at 15–35% pick-up. Its limits are penetration and uniformity control, not chemistry.

6. Comparison — the honest balance sheet

ROUTE	WATER USE	ENERGY PATTERN	EFFLUENT	COST / MATURITY
Conventional aqueous exhaust	High (LR 1:8–1:20 typical)	Bath heating + full drying	Coloured, saline; needs treatment	Lowest capital; universal fibres/dyes
Chlorinated-solvent dyeing	None in bath	Fast solvent evaporation; distillation load	No bath effluent, but solvent emissions risk	High capital; declined commercially
Supercritical CO_2	None in dyeing	Compression + heating; no water drying	Essentially none from dyeing; dye recovered	High-pressure capital; commercial for polyester
Foam dyeing	Very low (15–35% pick-up)	Drying energy cut ~ up to two-thirds (Topic 1)	Small volume, concentrated	Moderate retrofit; control-sensitive
Digital inkjet coloration	Low (treatment + wash-off remain for dye inks)	Steaming + washing for dye inks; low for pigment	Small; no screen/paste waste	High ink cost; ideal short runs/patterns
Ultrasound / microwave / plasma assist	Similar bath, better used	Shorter / lower-temperature cycles claimed	Similar, sometimes less chemical	Mostly pilot/niche; scale-up is the issue

Read the table as a portfolio, not a contest: a mill may run low-liquor aqueous jets for cotton, $scCO_2$ or conventional HT for polyester, foam for finishing and inkjet for patterned short runs — each where its balance sheet wins.

7. Glossary — key terms

- **Perchloroethylene:** chlorinated solvent historically used for solvent dyeing and dry cleaning.
- **Supercritical fluid:** substance above its critical temperature and pressure, with liquid-like density and gas-like transport.

- **Critical point (CO_2):** 31.1 °C and 73.8 bar.
- **Cavitation:** formation and collapse of micro-bubbles under ultrasound, producing local micro-jets.
- **Air-flow dyeing:** dyeing in which an air stream transports fabric/liquor, allowing very low liquor ratios.

8. Key points

- Solvent dyeing worked technically but declined on cost, recovery losses and chlorinated-solvent regulation.
- $scCO_2$ (T_c 31.1 °C, P_c 73.8 bar; dyeing typically ~120 °C, ~250 bar) keeps the waterless idea with a benign, recyclable medium — proven for polyester.
- Modern assists (foam, inkjet, ultrasound, microwave, plasma) each cut a specific cost; none is universal.
- Judge every route on the same four accounts: water, energy, effluent, cost.

9. Summary

Solvent dyeing proved that colour can be delivered without water — and its decline proved that the medium's cost, safety and recovery decide commercial survival. Supercritical CO_2 applies that lesson with a non-toxic, fully looped medium and dry fabric out, currently strongest for polyester. Around these, foam dyeing, air-assisted low-liquor dyeing, digital inkjet coloration and ultrasound/microwave/plasma assistance each modernise a part of the aqueous process. The engineer's task is matching route to fibre, dye class and run length on verified water, energy, effluent and cost figures.

10. Practice

1. State the critical temperature and pressure of CO_2 and explain in two sentences why solvent power rises with pressure above the critical point.
2. Give three distinct reasons chlorinated-solvent dyeing declined, one each from cost, operations and regulation.
3. A mill dyes only cotton with reactive dyes. Explain why neither historical solvent dyeing nor standard $scCO_2$ dyeing serves it directly, and name two modern routes from this sheet that do help it.
4. In the $scCO_2$ loop, what happens in the separator, and why does depressurisation cause it?
5. Why is "air dyeing" more accurately described as air-assisted low-liquor dyeing?

Answer key

1. T_c = 31.1 °C, P_c = 73.8 bar. Above the critical point, raising pressure raises fluid density towards liquid-like values; higher density means more dye molecules can be solvated per unit volume, so solvent power increases.
2. Cost: sealed machines, condensers and distillation plant plus solvent make-up. Operations: recovery losses at loading/unloading/maintenance broke the near-total recycle the economics required. Regulation/health: exposure limits and contamination risk for chlorinated solvents such as perchloroethylene raised compliance cost and liability.
3. Reactive dyeing needs water and alkali for ionisation, diffusion and covalent fixation — a chlorinated solvent cannot host that chemistry; standard scCO_2 dissolves non-polar disperse dyes, and cotton has no matching CO_2-soluble dye system in commercial use. Helpful routes: foam dyeing/finishing (low pick-up) and digital inkjet with reactive inks; also low-liquor air-flow exhaust dyeing.
4. As CO_2 expands and its density falls, its solvent power collapses; dissolved unused dye precipitates and is collected for reuse, while clean CO_2 passes on to be condensed and recycled.
5. Because the dye is still carried and fixed from a small aqueous phase — air provides transport and circulation (and allows very low liquor ratio), it does not dissolve or deliver the dye as a true solvent would.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib